Ahmedabad University

CSE400: Fundamentals of Probability in Computer science

Section 1

Computer Networks

Group 16

Milestone 2

Optimizing UAV Deployment for Multi-UAV Cooperative Sensing

Project Context and Motivation

Title: Optimizing UAV Deployment for Multi-UAV Cooperative Sensing

Domain: Computer Networks

Real-world Context:

- Multiple UAVs collect signals from environment
- Signals are corrupted by noise and interference
- System must detect whether a target signal exis

Why uncertainty is intrinsic?

- Presence of random noise (AWGN)
- Unknown signal strength and environment
- Variability in wireless channel and UAV sensing

Project Objective and Scope

Detection modeled as Binary Hypothesis Testing

$$H0 : y(k) = w(k)$$
$$H1 : y(k) = x(k)+w(k)$$

Where:
- y(k) = Received signal at UAV
- w(k) = Additive white Gaussian noise (AWGN)
- x(k) = Target signal

Objective:
- Design eigenvalue-based probabilistic detector
- Determine presence or absence of target signal
- Achieve high probability of detection Pd
- Maintain controlled false alarm probability Pfa

What is decided?
- Whether signal is present or absent
- Based on noisy UAV observations

Project System Overview

End-to-End Pipeline

- UAV collects received signal samples
- Covariance matrix is computed from samples
- Eigenvalues are extracted
- Test statistic is formed and compared with threshold
- Each UAV makes a local decision
- All UAV decisions are combined to produce the final detection

Major Components

- Signal sensing
- Eigenvalue-based detection
- Threshold decision
- Cooperative decision combining

Sources of Uncertainty in the Project

Measurement Noise

- w(k) is additive white Gaussian noise (AWGN)

Signal presence uncertainty

H0 : y(k) = w(k)
H1 : y(k) = x(k)+w(k)

Random test statistic

- Received signal contains noise and uncertainty
- Test statistic becomes a random variable

Gaussian approximation (for large K >> L)

$$H_0 : T \sim \mathcal{N}\left(1, \frac{2}{K}\right)$$

$$H_1 : T \sim \mathcal{N}\left(1 + SNR, \frac{2(1+SNR)^2}{K}\right)$$

Key Random Variables and Their Roles

Received signal y(k)

Noise w(k)

Target signal x(k)

Sample covariance matrix Ry

$$R_y = \frac{1}{K}\sum_{k=0}^{K-1} y(k)\, y^H(k)$$

Eigenvalues λ1,…,λL

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_L$$

Test statistic T

$$T = \frac{\lambda_1}{\lambda_L}$$

Probability of false alarm Pfa

$$P_{fa} = P(T \geq \lambda_{EBD} \mid H_0)$$

Probability of detection Pd

$$P_d = P(T \geq \lambda_{EBD} \mid H_1)$$

Detection threshold λEBD

$$\lambda_{EBD} = 1 + \sqrt{\frac{2}{K}}\, Q^{-1}(P_{fa})$$

Gaussian approximation of T (for K >> L)

$$H_1 : \; T \sim \mathcal{N}\left(1 + SNR, \frac{2(1+SNR)^2}{K}\right)$$

$$H_0 : \; T \sim \mathcal{N}\left(1, \frac{2}{K}\right)$$

Probabilistic Models and Assumptions

Joint Detection Probability

$$P_{coop} = 1 - \prod_{u=1}^{U} (1 - P_d^u)$$

Key Idea

- System detects target if any UAV detects
- Focus is on avoiding missed detection
- More UAVs ⇒ higher reliability

Importance

- Enhances overall detection performance
- Reduces probability of missed detection
- Improves system robustness under noise

Applications

- Optimizing UAV placement and orientation
- Evaluating cooperative sensing systems
- Decision making under uncertain environments

Probabilistic Reasoning and Inference Logic

Sample covariance:

$$R_y = \frac{1}{K}\sum_{k=1}^{K} y(k)\, y^H(k)$$

Test statistic:

$$T = \frac{\lambda_1}{\lambda_L}$$

False alarm:

$$P_{fa} = Q\left(\frac{\lambda_{EBD} - 1}{\sqrt{2/K}}\right)$$

Threshold:

$$\lambda_{EBD} = 1 + \sqrt{\frac{2}{K}}\, Q^{-1}(P_{fa})$$

Detection probability:

$$P_d = Q\left(Q^{-1}(P_{fa}) - \frac{SNR \cdot \sqrt{K/2}}{1 + SNR}\right)$$

Operationalizing Probabilistic Reasoning in the Project

Operationalizing Probabilistic Reasoning

- Implemented using simulations
- Compare random vs optimized UAV deployment

Simulation Approach

- Generate signal and noise samples
- Perform eigenvalue-based detection
- Compute detection probabilities

Evaluation Metrics

- Cooperative detection probability
- False alarm rate
- Optimization performance

Simplifications

- Simplified noise and channel models
- Focus on core probabilistic behavior

Current Limitations and Conceptual Gaps

Modeling Limitations (Appendix Context)

- Gaussian approximation valid only for large sample size KKK
- Assumes eigenvalue ratio follows normal distribution
- Noise modeled as ideal AWGN
- Real-world noise may not be Gaussian

Assumption Gaps

- Independence between UAV observations assumed
- Fixed UAV height and static targets
- Perfect communication with fusion center

Optimization Challenges

- PSO computational complexity increases with UAV count
- Convergence not guaranteed for all scenarios

PDF and CDF

```
========================================
 Milestone-2 Practical Simulation Started
========================================
Samples K = 1000
Sensors L = 10
SNR(dB)   = 0
Runs N    = 10000
----------------------------------------
Simulation Completed Successfully!
```

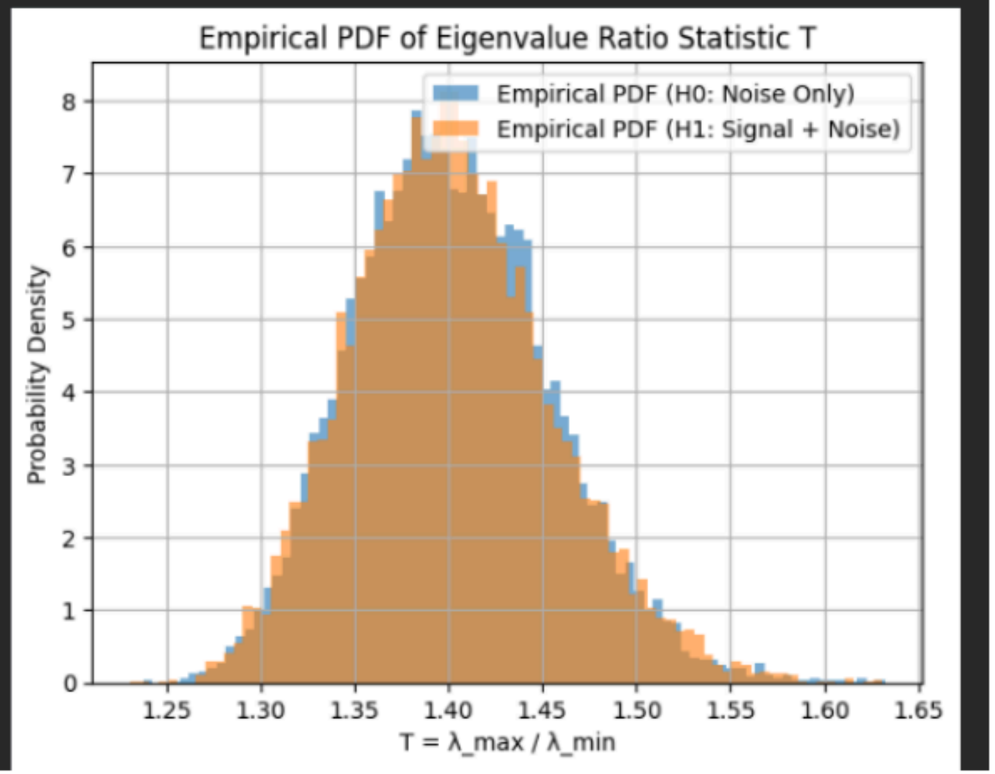

Figure 1: Empirical PDF under H_0 and H_1.

This simulation provides empirical intuition about the separation between the distributions of the test statistic under H_0 and H_1, motivating the theoretical derivation of P_{fa} and P_d.

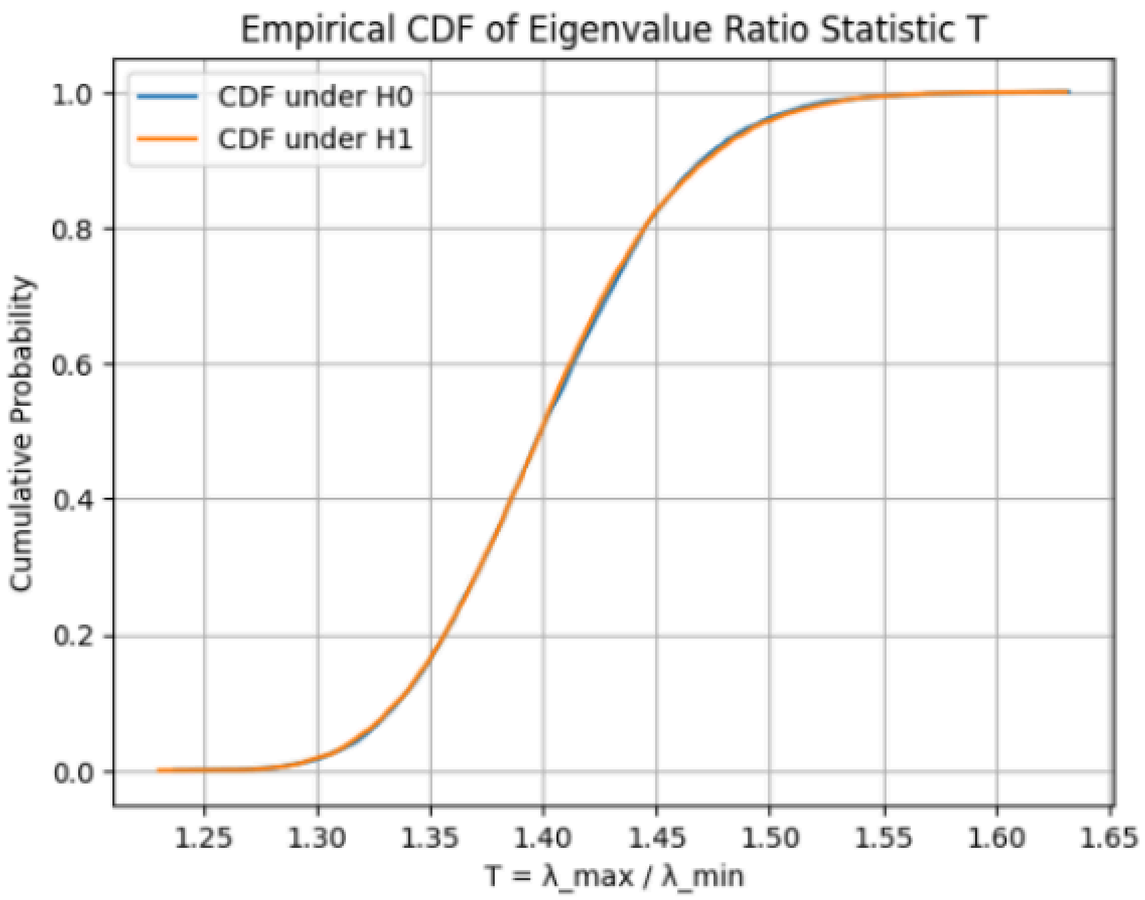

Figure 2: Empirical CDF under H_0 and H_1.

Planned Refinements and Role Coordination

Future Refinements

- Work on Mathematical Modelling as a whole again with acute detail.
- Improve noise and channel modeling.
- Validate probabilistic assumptions.
- Extend to larger UAV networks.
- Enhance optimization algorithms.

Role Coordination

- Code development and simulations for PDF and CDF
- Model validation
- GitHub for version control
- Regular team coordination

Summary of Group-Level Understanding

Group-Level Understanding

- Derived eigenvalue-based detection statistics
- Understood role of covariance matrix and eigenvalues
- Understood the level of mathematical modelling needed in the next milestone.

Strengths

- Strong understanding of mathematical derivations
- Ability to implement detection and simulation framework

Conceptual Gaps

- Limited focus on probabilistic modeling aspects
- Incomplete understanding of distribution assumptions
- Need deeper interpretation of uncertainty propagation